NLU course project: Intent Classification and Slot Filling

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1. Introduction

This project compares two strategies for joint intent classification and slot filling on ATIS, evaluated with Intent Accuracy and chunk-level Slot F1 (CoNLL): training a Transformer *from scratch* (Part A) versus fine-tuning pre-trained encoders and decoders (Part B). For Part A, a GPT-2-style decoder-only Transformer is the chosen architecture, with a greedy hyper-parameter search performed over learning rate, architecture, and dropout. For Part B, pre-trained BERT [1] and GPT-2 [2] are used, adapting the sub-tokenization strategy of Chen et al. [3] for slot label alignment. The best model overall is BERT-base, reaching test Slot F1 0.955 and Intent Accuracy 0.975, against 0.888 and 0.934 for the best from-scratch model: pre-trained bidirectional representations give a clear advantage on both tasks.

2. Implementation details

The backbone architectures (GPT-2 from scratch and the fine-tuning setup) follow the course lab; the contributions below are original to this work.

CLS placement (Part A). Since the model is causal, the `CLS` token is appended at the *end* of each utterance rather than at the start. The last position is the only one that has attended to the full sequence, so its hidden state is used for intent classification; slot decoding excludes this extra position before scoring with CoNLL.

Per-architecture intent representation (Part B). BERT uses the `[CLS]` hidden state at position 0. GPT-2 is causal, so intent classification instead uses the hidden state at the last *real* token, retrieved via the per-sequence length rather than the last padded position.

Sub-token label alignment (Part B). Following the strategy discussed in [3], only the first sub-token of each word carries the true slot label; remaining sub-tokens and special tokens are excluded from the slot loss via `ignore_index`. This reference is used as conceptual guidance for the alignment rule, not as a code source.

Multi-seed greedy search. Each candidate in the greedy hyper-parameter search (learning rate first, then architecture, then dropout) is run over multiple seeds *before* being compared to the incumbent, rather than only the final model. This reduces the risk of selecting a configuration that owes its apparent advantage to a favorable seed instead of a real effect of the hyper-parameter.

3. Results

All models are evaluated on the test set over multiple seeds (mean±std); hyper-parameters are selected on the *dev* Slot F1 to avoid test-set leakage. Table 1 and Figure 1 report Part A: the learning-rate search already reaches most of the achievable performance, and only `n_heads` contributes a further, modest gain – every other architecture candidate falls below the incumbent and is discarded. Table 2 reports Part B: BERT clearly outperforms GPT-2 on slot filling (0.955 vs. 0.915 test F1), consistent with BERT's bidirectional context being better suited to per-token tagging than GPT-2's left-to-right attention. Scaling to the large/medium variant does not improve test metrics for either family despite 3× more parameters, suggesting ATIS is already saturated by base-sized pre-trained models. Figure 2 makes this explicit: moving from a from-scratch model to a pre-trained one yields a large jump in Slot F1, while further scaling within Part B yields essentially none.

Table 1: *Part A — best model per step (231K param. throughout). Final: d=64, 4 heads, 2 layers, FFN 256, dropout 0.*

Step	Adopted	Dev F1	Test Slot F1	Test Int. Acc
0 – lr	0.01	0.929	0.876±0.007	0.921±0.017
1 – architecture	heads=4	**0.941**	**0.888±0.006**	**0.934±0.006**
2 – dropout	none (0)	0.941	0.888±0.006	0.934±0.006

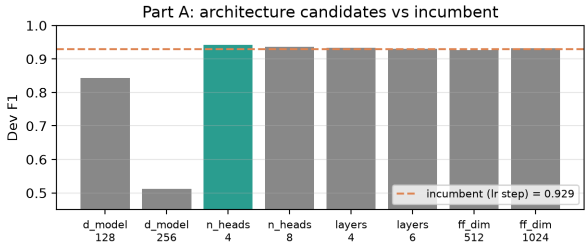

Figure 1: *Part A: dev F1 of every architecture candidate against the incumbent score (dashed line); only `n_heads`=4 wins.*

Error analysis. Beyond aggregate scores, Figure 3 breaks down test Slot F1 by slot type for the two best fine-tuned models. On the most frequent slots both are strong, but BERT's advantage widens precisely on slots whose interpretation depends on context from *both* sides – `airline_name` (0.99 vs. 0.86), `depart_time.period_of_day`, `aircraft_code`, `class_type` – confirming that bidirectional attention, not scale, drives the gap, since GPT-2-medium already has 3× BERT-base's parameters. The hardest slot for both is the generic `city_name` (F1 0.64/0.55), routinely confused with the directional `fromloc/toloc.city_name`. Intent classification is by contrast nearly solved: the best BERT misclassifies only 23 of 893 test utterances, and the residual errors (Figure 4) are dominated by compound intents such as `flight+airfare` collapsing onto their majority component `flight`, rather than genuine semantic confusions.

4. References

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language under-

Table 2: *Part B — fine-tuning (best lr/dropout per family).*

Model	Param	Dev F1	Test Slot F1	Test Int. Acc
BERT-base	109.6M	**0.984**	**0.955±0.002**	**0.975±0.001**
BERT-large	335.3M	0.983	0.955±0.001	0.975±0.002
GPT-2-base	124.6M	0.963	0.915±0.004	0.971±0.002
GPT-2-medium	355.0M	0.964	0.913±0.002	0.969±0.003

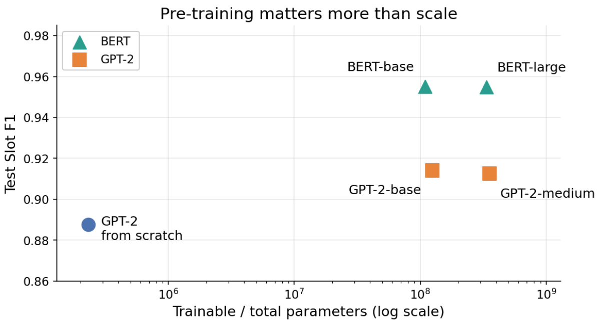

Figure 2: *Test Slot F1 vs. parameter count (log scale): the jump from Part A to Part B dwarfs any gain from scaling within Part B.*

standing," in *Proceedings of NAACL-HLT*, 2019.

[2] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," OpenAI, Tech. Rep., 2019.

[3] Q. Chen, Z. Zhuo, and W. Wang, "BERT for joint intent classification and slot filling," in *arXiv preprint arXiv:1902.10909*, 2019.

Declaration of AI use: a generative-AI assistant (Claude, Anthropic) was used to help set up and debug the remote GPU training environment, run the experiments, generate the figures, and edit the language of this report. The choice of hyper-parameters to search, the model architecture, and the interpretation of the results are the author's own.

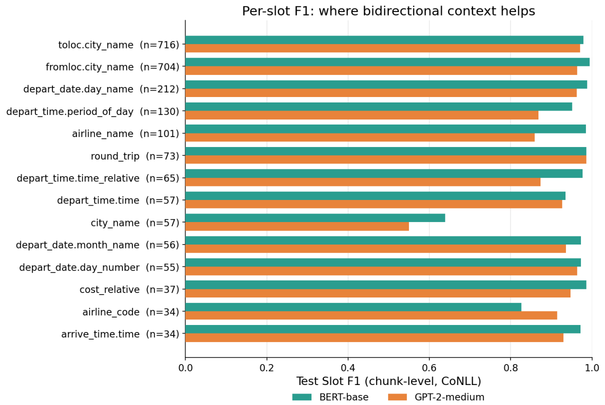

Figure 3: *Per-slot test Slot F1 (slots with support $\geq$ 20, n = test occurrences): BERT vs. GPT-2-medium. The gap widens on context-dependent slots, not on the most frequent ones.*

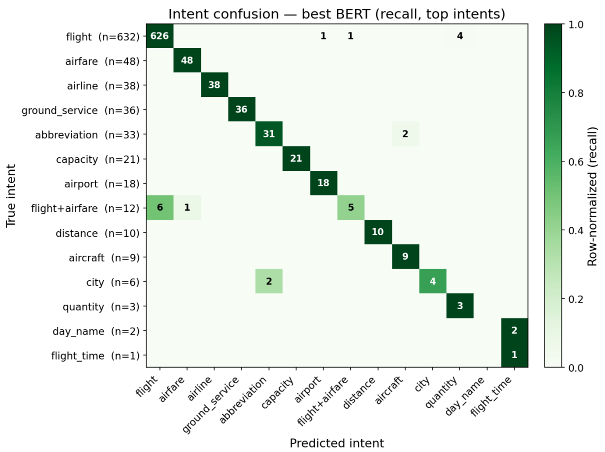

Figure 4: *Intent confusion (row-normalized recall) of the best BERT on the ATIS test set; cells show raw counts. Only 23/893 errors, mostly compound intents mapped to* `flight`.